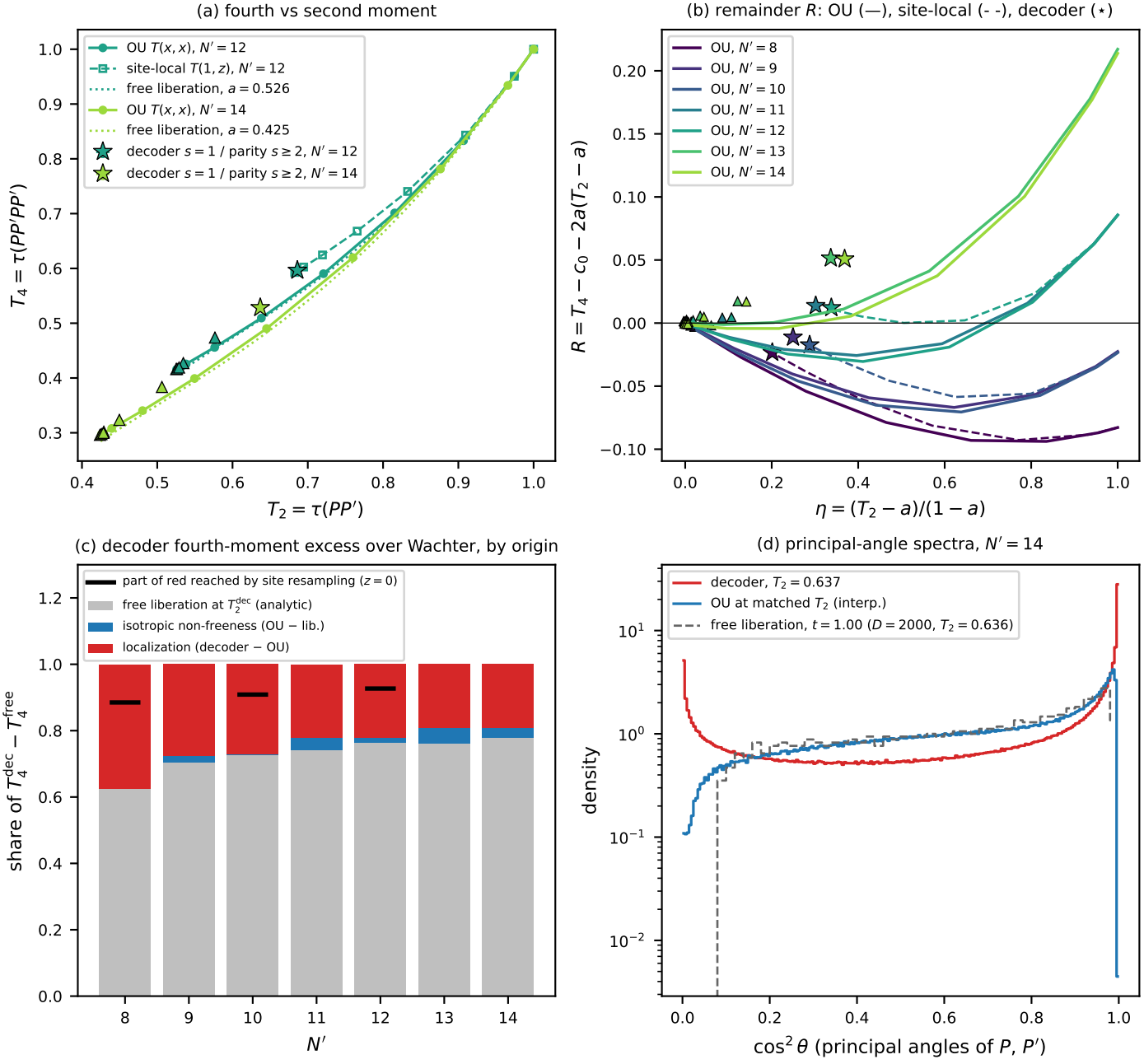
(a) fourth vs second moment
OU $T(x,x)$, $N'=12$
site-local $T(1,z)$, $N'=12$
free liberation, $a=0.526$
OU $T(x,x)$, $N'=14$
free liberation, $a=0.425$
decoder $s=1$ / parity $s\geq 2$, $N'=12$
decoder $s=1$ / parity $s\geq 2$, $N'=14$
$T_4=\tau(PP'PP')$
$T_2=\tau(PP')$
(b) remainder R: OU (—), site-local (- -), decoder ($\star$)
OU, $N'=8$
OU, $N'=9$
OU, $N'=10$
OU, $N'=11$
OU, $N'=12$
OU, $N'=13$
OU, $N'=14$
$R=T_4-c_0-2a(T_2-a)$
$\eta=(T_2-a)/(1-a)$
(c) decoder fourth-moment excess over Wachter, by origin
part of red reached by site resampling ($z=0$)
free liberation at T_2^{dec} (analytic)
isotropic non-freeness (OU − lib.)
localization (decoder − OU)
share of $T_4^{\mathrm{dec}}-T_4^{\mathrm{free}}$
N'
(d) principal-angle spectra, $N'=14$
decoder, $T_2=0.637$
OU at matched T_2 (interp.)
free liberation, $t=1.00$ ($D=2000$, $T_2=0.636$)
density
$\cos^2\theta$ (principal angles of P, P')